Validity Analysis: ASAP_PLUS_PLUS

Target context: Qatari and Arab High School Arabic Essay Writers (Grades 10–12)

Overall risk: **HIGH**

Deployment Context

Use case: Support high school students (G10-12) in editing and revising their Arabic essay drafts before submitting them for grading.

Target population: high school students (G10-12) in Arab countries (especially Qatar)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	1	Serious concern	high
Input Content ⚠	1	Serious concern	high
Input Form ⚠	1	Serious concern	high
Output Ontology ⚠	1	Serious concern	high
Output Content ⚠	1	Serious concern	high
Output Form ⚠	1	Serious concern	high
Average	**1.0**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

ASAP++ is severely misaligned with the target deployment across all six validity dimensions, all of which were flagged HIGH priority in the elicitation. The benchmark provides English-only essays from native English-speaking US Grade 7–10 students, scored on US-pedagogy-derived genres (argumentative, source-dependent, narrative) by India-based English-MA annotators anchored to US holistic scores, with numeric ordinal output evaluated by QWK. The deployment requires Arabic (MSA) Grade 10–12 essays under Qatar MOEHE and other Arab national curricula, covering literary analysis, religious-text commentary, and cultural commentary genres absent from the benchmark, evaluated against Arab curriculum stakeholders, and producing natural-language revision suggestions with explanatory rationale. There is no overlap in language, script, age band, curriculum, annotator expertise, genre coverage, or output modality. ASAP++ is at best a methodological reference (multi-trait scoring structure, QWK reliability template) for a future Arabic AES effort; it cannot serve as a validity anchor for the deployment. More directly relevant Arabic resources — LAILA (7,859 high school essays, seven-trait rubric, QRDI-funded), QAES/QCAW (195 Qatari argumentative essays), and Arwi (Arabic feedback-generation system) — should replace ASAP++ as the primary references.

Practical Guidance

What This Benchmark Measures

ASAP++ measures the ability to predict numeric ordinal trait scores (content, organization, word choice, sentence fluency, conventions, and analogous traits) for English-language US student essays in argumentative, source-dependent, and narrative genres, evaluated against US-anchored ground truth via QWK. None of the strongest dimensions of the benchmark align with the target deployment; all six dimensions score 1.

Construct Depth

Within its English/US scope, the benchmark provides moderate construct depth on multi-trait holistic scoring with reasonable inter-rater methodology (QWK-anchored QC). For the Arab high school Arabic writing construct, however, depth is effectively zero: the benchmark probes none of the rhetorical, morphological, diglossic, or curricular dimensions that define writing quality in MSA, and produces no evidence about feedback-generation quality.

What Else You Need

Wholesale supplementation is required. Replace ASAP++ as the validity anchor with: (1) LAILA [WEB-12] for trait taxonomy and high school Arabic essay coverage, (2) QAES/QCAW [WEB-3] for Qatari-anchored argumentative writing, (3) Arwi [WEB-22] for Arabic feedback-generation methodology, and (4) recruit Qatar MOEHE-curriculum teacher annotators for genre-specific rubric validation across literary analysis, religious-text commentary, and cultural commentary essays. Build an Arabic NLP pipeline on CAMeL Tools / CAMeLBERT-MSA [WEB-7, WEB-9] rather than Stanford CoreNLP. Commission a feedback-generation evaluation framework (rubric-aligned human evaluation with Qatari teachers) since QWK cannot validate generated revision suggestions.

ID	Evidence
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)
WEB-3	QAES annotated Qatari argumentative writing with trait-specific conventions appropriate to Qatari context (https://www.mdpi.com/2306-5729/10/9/148)
WEB-22	Arwi (CAMeLBERT-MSA, ZAEBUC) demonstrates the kind of feedback-generation output form the deployment requires; QWK is insufficient to evaluate such systems (https://arxiv.org/html/2504.11814v1)
WEB-7	CAMeL Tools provides MSA morphological analysis, POS, dialect ID and diacritization — the Arabic counterpart ecosystem to Stanford CoreNLP (https://aclanthology.org/2020.lrec-1.868.pdf)
WEB-9	CAMeLBERT-MSA is the standard pre-trained Arabic encoder used in ZAEBUC, LAILA, and TAQEEM AES systems (https://nyuad.nyu.edu/en/research/faculty-labs-and-projects/computational-approaches-to-modeling-language-lab/resources.html)